FELIX

48-Hour Engineering Systems Report

From agent harness to governed autonomous engineering institution

Canonical code window

31ed8a8ebd28 -> c6989dc5168f

September 22-24, 2026

Bottom line

In 48 hours, Felix sustained a high-concurrency self-hosted development burst while growing verification, authority semantics, continuity, and operational controls nearly as aggressively as implementation. The dominant new problem was no longer code generation: it became evidence freshness, candidate reconciliation/landing, organizational observability, and routing scarce human attention.

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1. Executive summary

The 48-hour period was not primarily a feature sprint. It was a systems-maturation event. Felix ran at a scale that exposed organizational failures quickly, then converted many of those failures into durable mechanisms: lifecycle telemetry, handoff ownership, schema enforcement, selective evidence invalidation, bounded unattended operation, human authority boundaries, stronger deny semantics, self-audit, and stranger-project portability.

Metric	Measured value	Interpretation
Raw model token traffic	6.975B	Scale indicator only; 97%+ was cache-read traffic, so this is not equivalent to fresh compute.
Unique model responses	30,694	Large execution volume across parent and subagent contexts.
Tool calls	32,187	Execution was tool-heavy; Bash alone accounted for 26,295 calls (~81.7%).
Average concurrency	12+ sessions	The working mode had already become multi-session rather than serial.
Landed commits	50	Canonical mainline output in the frozen code window.
Gross line churn	31,001	24,072 additions and 6,929 deletions; handoff migration materially distorts raw LOC.
Engine vs tests	+5,995 vs +6,060	Verification surface expanded roughly in step with engine code.
Felix stop holds	363	Actual enforcement events when a current passing gate was absent.
Median full-suite runtime	9m 56s	Verification is now a material latency

Metric	Measured value	Interpretation
		component, not a free afterthought.
Resume share	37.6%	Recorded session activations increasingly reuse existing engineering threads.
Sessions <=1 human text turn	81.6%	Low textual shepherding, but UI clicks that spawn Suggested Tasks are not counted.

Core interpretation

Felix is no longer best modeled as a coding accelerator. The evidence fits a human-directed, agent-executed, self-hosted engineering organization in which the central technical problem is maintaining trustworthy canonical state while execution capacity scales faster than integration capacity.

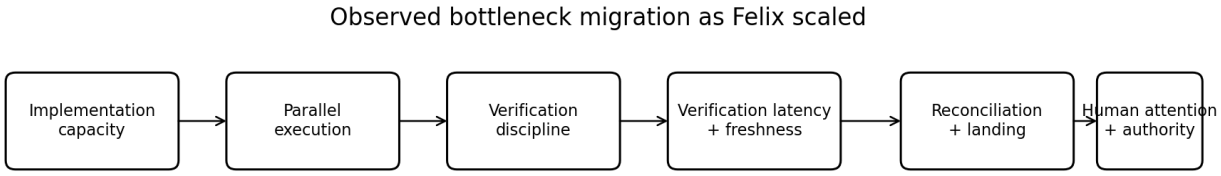


Figure 1. Observed bottleneck migration during the burst.

2. Measurement boundary and methodology

The canonical code window is frozen at 31ed8a8ebd28be9cdf0c7dbf7708adabb767eca4 -> c6989dc5168f5113745591fde1115b7270dfe933. This window contains exactly 50 landed commits and is the basis for statements about accepted code changes. Operational telemetry (audit, lifecycle, human-turn analysis, test runtimes) was sampled around the same period, sometimes slightly after the end SHA while main was still moving. The report therefore separates code-window facts from near-window operating measurements rather than pretending all counters share a perfectly synchronized cutoff.

The report combines five evidence classes: Git history and diffs; Claude Code JSONL telemetry; Felix audit output; Felix lifecycle logs; and controlled full-suite runtime measurements. External-project worktrees are treated as in-flight execution evidence, not as accepted output. PR throughput and latency are reconstructed from the PR chronology and are therefore approximate and right-censored by PRs still open at the observation point.

Measurement discipline

Whenever a metric can be over-read, the report uses the narrower claim. Raw token traffic is not fresh compute. Human text turns are not total human actions. Parent-session audit results are not organization-wide verification rates. Branch inventory is not accepted output. In-flight worktrees are not landed productivity.

3. Scale: compute, parallelism, and execution

Felix operated at a scale where coordination effects are visible rather than hypothetical. The 48-hour telemetry contained 30,694 deduplicated model responses and 32,187 tool calls. Raw token traffic was 6.975B, of which 6.788B was cache read, 156.237M cache creation, 30.404M output, and only ~69.9K uncached input. Cache read therefore dominates the traffic profile. The system repeatedly reused long, stable context rather than paying the full cost of reconstructing institutional/project context on every turn.

Subagents represented 3.728B tokens versus 3.247B for parent sessions, or 53.45% of token traffic. This matters twice: first, it confirms that the execution topology was already distributed; second, it exposes a major observability gap because the current Felix audit explicitly does not read subagent transcripts.

Layer	Traffic / count	Interpretation
Parent sessions	3.247B raw tokens	Human-facing coordination and primary session execution.
Subagents	3.728B raw tokens (53.45%)	More than half of raw traffic occurred below the parent-session layer.
Unique responses	30,694	High response volume after response-ID deduplication.
Tool calls	32,187	Tool execution slightly exceeded response count.
Bash	26,295 tool calls	~81.7% of all tool calls; the operating substrate is heavily shell/process oriented.

The practical consequence is that Felix was already behaving less like one agent and more like a small engineering organization. The user reported averaging 12+ sessions operating concurrently, while worktree and branch inventory confirmed many independent execution contexts. The important question is therefore no longer whether agents can work in parallel; it is whether the system can govern, reconcile, and land that parallel work without exploding human attention or trust debt.

4. Human supervisory leverage

Across 445 parent transcript files, 190 active parent sessions were observed in the 48-hour window. Those sessions contained 471 human text turns, 18,946 assistant turns, and 8,354 parent-session tool calls. That yields 40.2 assistant turns and 17.7 tool calls per human text turn.

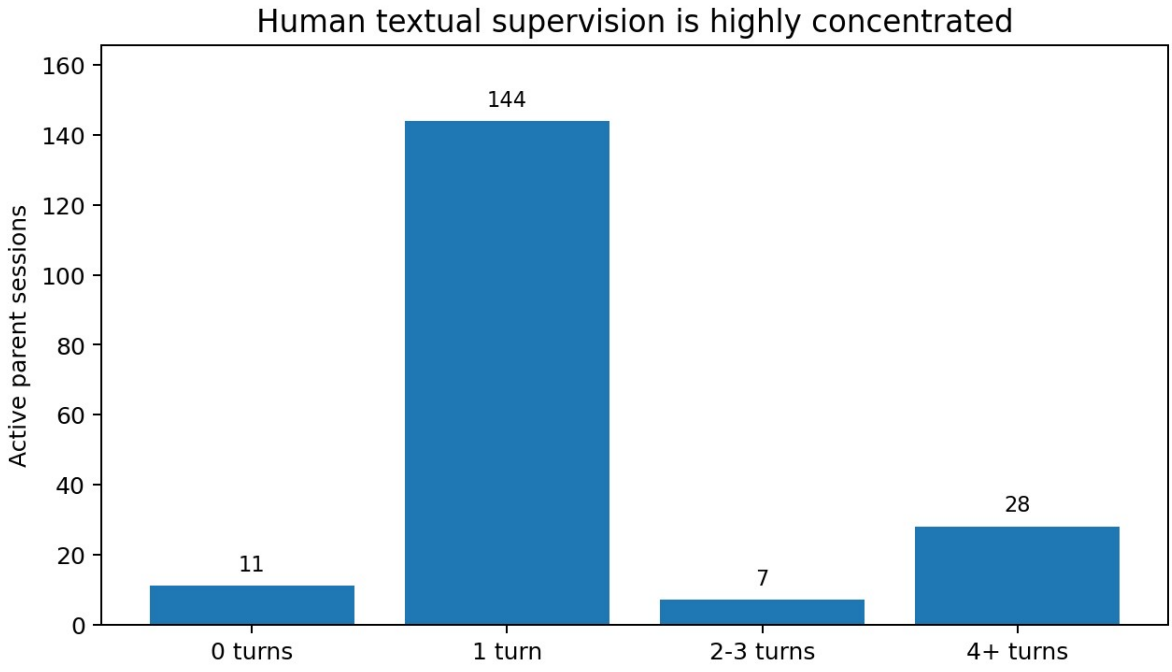


Figure 2. Distribution of human text turns per active parent session.

The distribution is more informative than the average. 144 of 190 active sessions had exactly one human text turn, 11 had none, and only 28 had four or more. In aggregate, 81.6% of sessions had at most one human text turn and 85.3% had at most three. A small tail absorbed most supervision: the 25 most human-intensive sessions (13.2% of sessions) accounted for 300 of 471 human text turns, or 63.7%.

Important attribution correction

These numbers measure textual supervision, not all human intervention. During the burst, Claude frequently surfaced in-session Suggested task cards and the user manually clicked Start with worktree. Those UI actions are not represented as ordinary human text turns. A large share of fix-oriented sessions therefore followed this pattern: Claude discovers a follow-on task -> human reviews/authorizes it -> a new isolated worktree/session is created -> the agent executes largely autonomously -> Felix governs verification and completion.

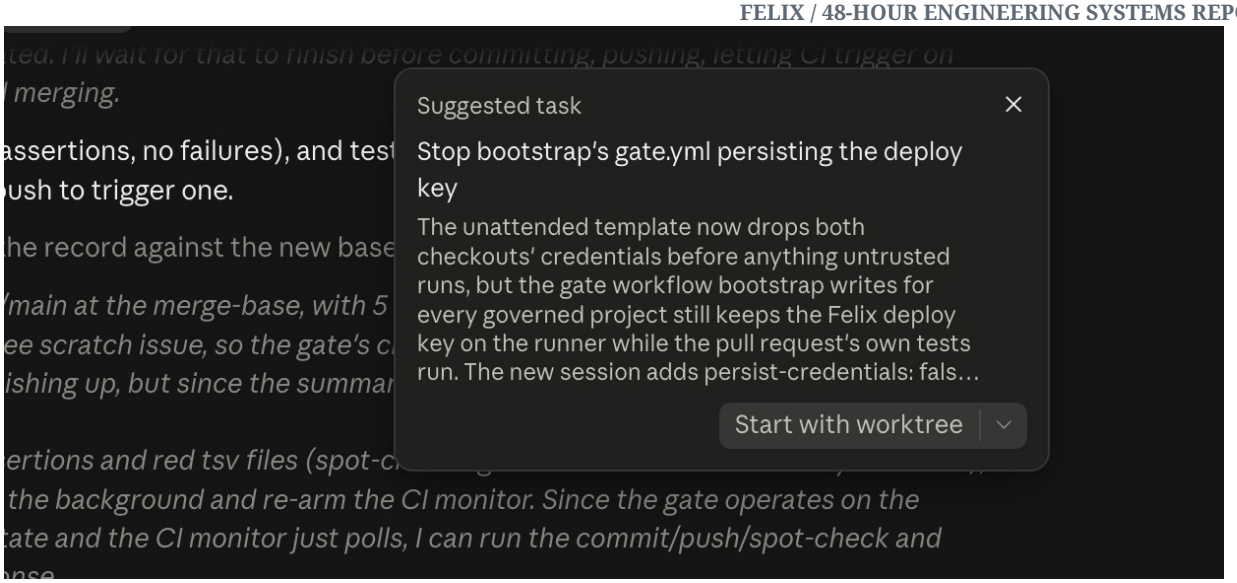


Figure 3. Example of Claude discovering a follow-on issue and proposing a one-click worktree task. This is task discovery plus human authorization, not Felix-autonomous spawning.

This distinction improves the architecture model. Claude is already providing useful local task discovery; the user still acts as a portfolio/authority filter; Felix governs the resulting execution. The human leverage signal is therefore real, but its cleanest interpretation is: once a task is authorized and spawned, most sessions require little textual shepherding.

5. Landed engineering output

Between the frozen base and end SHA, 162 files changed with 24,072 insertions and 6,929 deletions: 31,001 gross line touches and +17,143 net lines. Raw LOC is not a value metric, and this window contains a structural HANDOFF migration that inflates churn, so the more useful decomposition is by subsystem.

Category	Files	Additions	Deletions	Churn
Engine	41	5,995	379	6,374
Tests	3	6,060	106	6,166
Docs	6	1,780	0	1,780
HANDOFF monolith	1	234	6,433	6,667
Handoff state	72	7,308	0	7,308
Episode research	11	967	0	967
Other	26	1,728	11	1,739

The handoff subsystem alone contributed 13,975 lines of structural churn, much of it migration from one monolith to per-section state. Excluding that migration, engine plus tests dominate the normal engineering activity. The handoff subsystem still grew net, however: it was not merely file shuffling; it became a larger, more explicit institutional-memory surface.

6. Verification growth and quality coupling

The strongest quality signal is not that tests exist; it is that the verification surface scaled almost one-for-one with engine growth. Engine code added 5,995 lines while tests added 6,060. assertions.tsv grew from roughly 2,735 to 3,485 rows (+27.4%), and the test runner grew from roughly 16,822 to 21,988 lines (+30.7%).

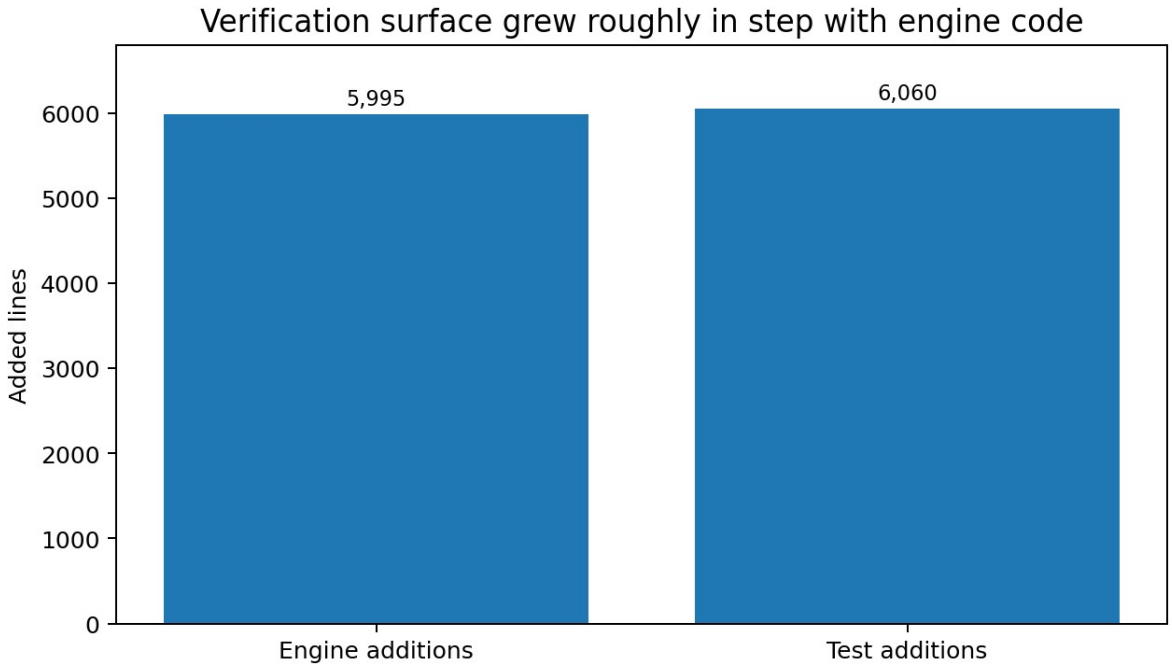


Figure 4. Engine and test additions were nearly equal during the frozen code window.

Verification was also coupled to accepted changes at the commit level. Of the 50 landed commits, 32 (64%) touched tests/run and 30 (60%) touched assertions.tsv. Every assertion-registry-touching commit in the measured set also touched tests/run. This is not code-coverage proof, but it is evidence that verification did not lag far behind implementation during the burst.

The distribution of edits inside tests/run was spread across many regions rather than one tiny hotspot. That reduces the case for a single localized contention point, but the file itself is now a ~22K-line shared orchestration artifact. Its size and centrality are architecture signals even if the current line-level edit distribution is not yet pathological.

7. Trust Plane in actual operation

The Felix audit supplied the most direct evidence that Trust Plane controls are economically active rather than decorative. Across 442 observable parent sessions, 154 meaningful stop-after-edit events were identified in 52 sessions. At those stops, 99 were nominally passed, 48 were stale, 7 had never run a recognized check, and 0 ended immediately after an explicitly failed recognized check.

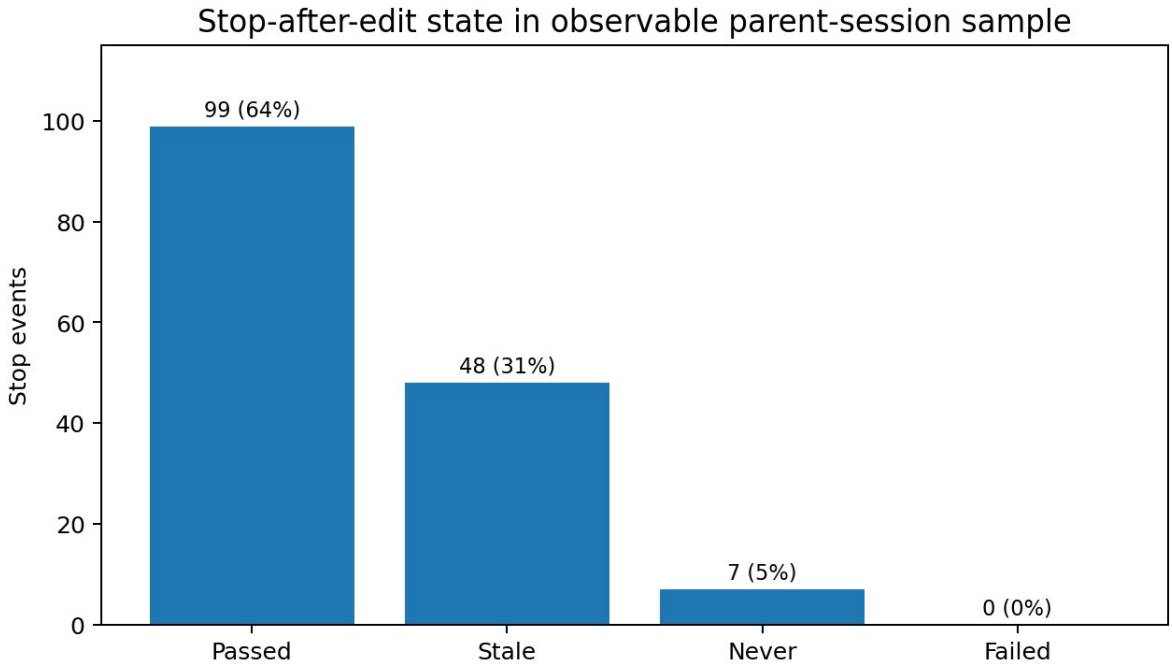


Figure 5. 35.7% of observed edit-ending stops lacked fresh recognized verification.

The combined stale-plus-never share was 35.7%. The failure mode was therefore not usually "test fails and the agent declares done." It was more subtle and more relevant to autonomous engineering: a check had run earlier but no longer corresponded to the current tree, or no recognized check had run at all. This is exactly the class of error that revision-bound evidence is meant to catch.

Felix held 363 stop attempts in 51 sessions because no current passing gate result existed for the tree. Of those, 113 coincided with turns that also appear in the stop-after-edit classification, 38 followed only scratch/config/memory edits, and 212 followed turns with no new edit. These holds demonstrate that Felix was repeatedly interposing itself between model-level completion and organization-level completion.

Trust Plane mechanism

The core question is not "did a test run at some point?" It is "for this tree, at this revision, do we possess acceptable evidence and the right authority to treat the work as complete?" The audit data shows why that distinction matters in practice.

The audit itself also exposed two observability defects. First, a malformed multibyte sequence originally caused the whole 459-transcript audit to abort until the locale was forced to byte-oriented handling. Second, subagents are invisible to the audit even though they represent 53.45% of raw token traffic. A Trust Plane whose own evidence ingestion can fail globally on one malformed source, or cannot see a majority of worker compute, has a clear next-generation observability agenda.

8. Institutional continuity

Lifecycle telemetry shows that session continuity is becoming operational rather than aspirational. Across governed projects, 102 lifecycle rows represented 64 project-local sessions: 53 startup events, 32 resumes, 16 handoffs, and 1 compact. Twenty-three sessions (35.9%) were resumed at least once and 12 (18.8%) spanned at least two calendar days. Resumes accounted for 37.6% of recorded session activations.

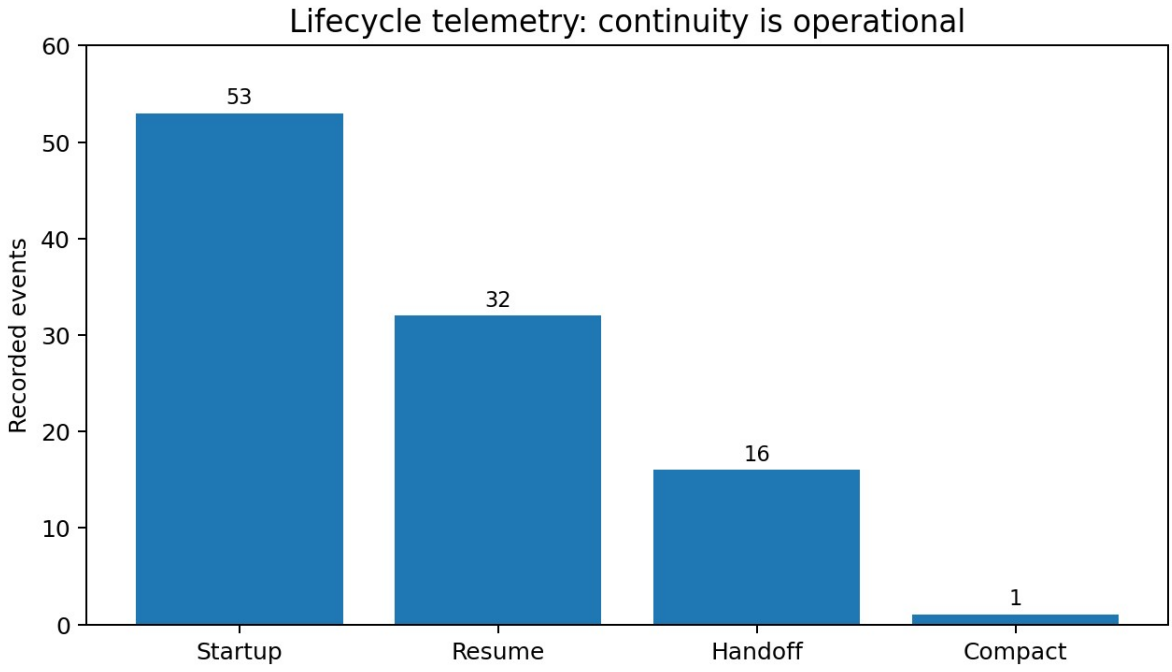


Figure 6. Lifecycle events across governed projects during the sampled period.

The logs show concrete continuity across context boundaries: sessions resume after interruptions and version changes; at least one ReserveGrid/Veldra session compacted and later resumed repeatedly; BehaviorChangeBench sessions also resumed across Sep 22-24. Because lifecycle instrumentation itself landed during the 48-hour window, these counts likely understate the early portion of the period.

The important remaining question is semantic continuity, not mere reopenability. A session that resumes but restores the wrong obligations, decisions, ownership, or revision assumptions has not preserved institutional continuity. Felix has crossed the first threshold - durable lifecycle state exists - but future measurement should test whether restored state is correct and sufficient.

9. Candidate generation and integration economics

The branch and PR data show the next bottleneck migrating upward. Seventy-seven branches were updated in the 48-hour period. Naively summing per-branch candidate commits produced 236; global deduplication reduced that to 175 unique non-main commits, meaning roughly 25.8% of branch-local candidate history was duplicated across branches. This is consistent with parallel architecture search, stacked work, and superseded candidates rather than a single linear queue.

The open-PR snapshot contained roughly 17 real PRs, including several conflicting or dirty candidates. One large stacked candidate represented ~38.6% of the pending additions by itself. The key observation is not the raw open count; it is that main was moving fast enough for candidates to age, require restacking, or lose freshness while they waited.

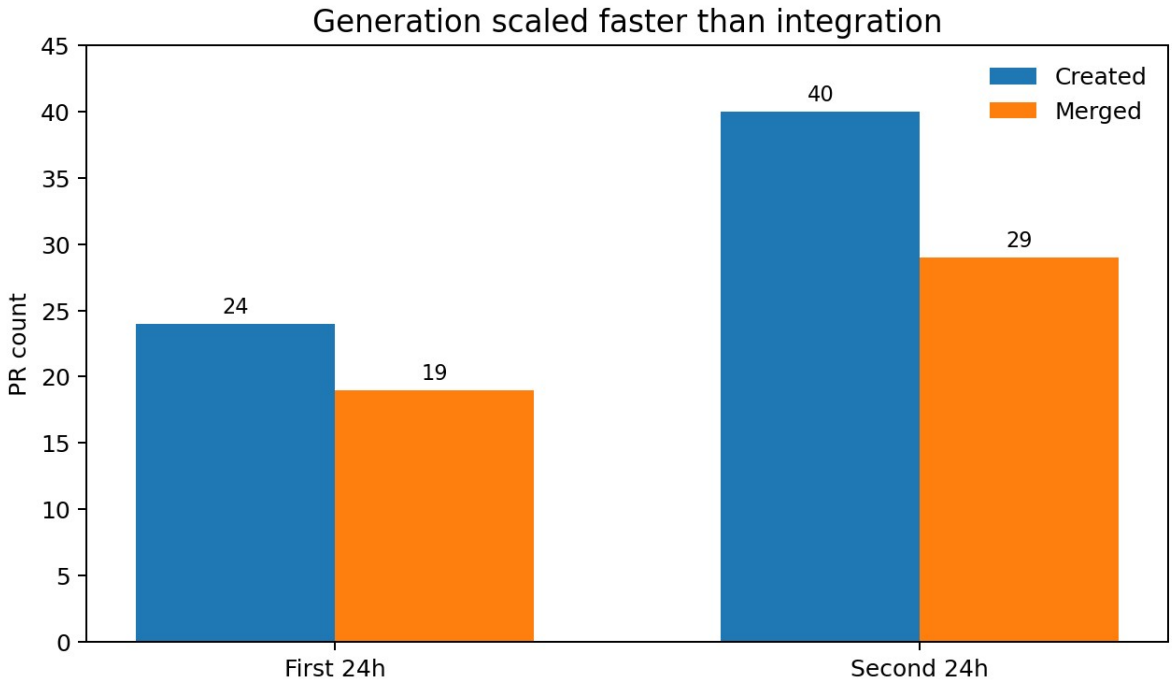


Figure 7. Reconstructed PR counts by 24-hour half. Generation accelerated faster than merging.

In the reconstructed chronology, PR creation rose from about 24 in the first 24 hours to about 40 in the second (+66.7%), while merges rose from about 19 to about 29 (+52.6%). Median create-to-merge time among merged PRs increased from ~0.55h to ~1.37h, with the upper tail worsening more sharply. These latency estimates are approximate and censored because slower PRs still open at observation time are not included in the merged sample.

Independent convergence

Felix itself diagnosed the same problem in its handoff: "no landing actor; readiness decays per merge." That qualitative self-diagnosis aligns with the external quantitative pattern: generation scaled faster than integration, so accepted-state maintenance became the limiting system.

10. Verification latency and evidence freshness

Three consecutive full-suite runs all exited 0 with 3,511 passed, 0 failed, and 3 skipped assertions. Wall times were 626.35s, 595.82s, and 556.85s. Mean wall time was 593.0s (9m53s), median 595.82s (9m56s), with ~5.9% coefficient of variation. The median cost was ~169.7 seconds per 1,000 passed assertions, and average throughput was ~5.93 passed assertions per second.

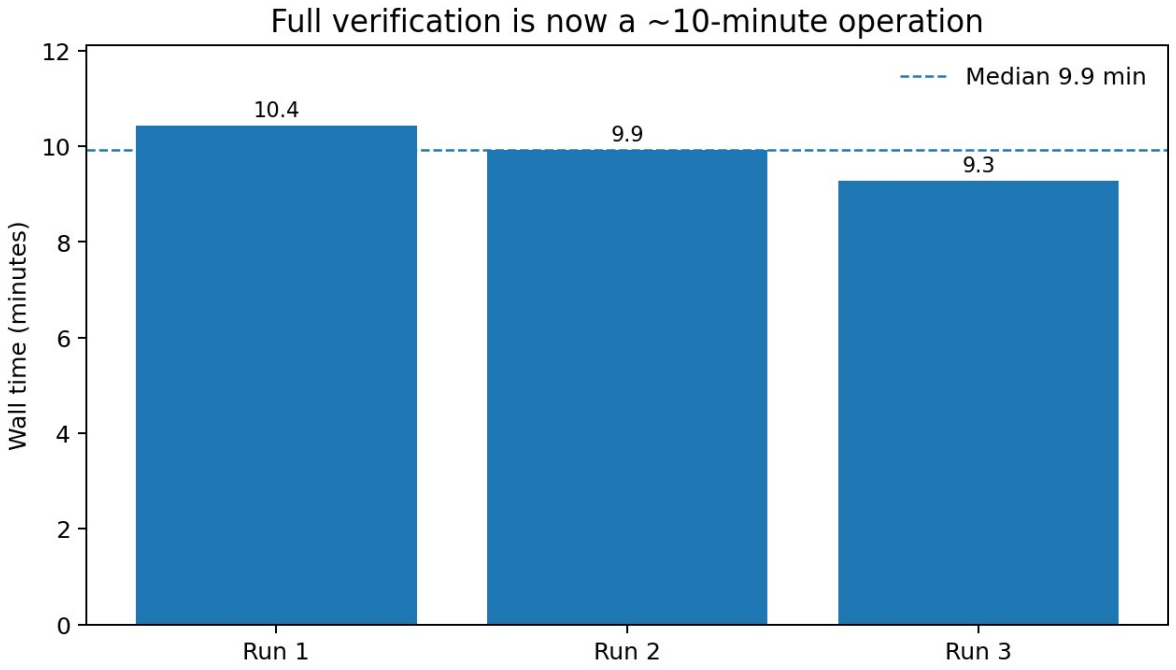


Figure 8. Three repeated full-suite measurements on Felix main.

The process-level CPU proxy, (user + sys) / real, averaged ~0.983. That does not prove the suite is wholly serial, but it shows little effective whole-run CPU parallel speedup. The runner is heavily shell/process/filesystem oriented, so wall time likely includes orchestration, sleeps/timeouts, process lifecycle, and fixture setup in addition to pure compute.

This matters because evidence acquisition itself now consumes roughly ten minutes. At 12+ concurrent sessions and a fast-moving main branch, a candidate can become stale during the act of qualifying itself. Verification latency therefore couples directly into integration economics: generation -> gate -> main advances -> neighboring candidates lose freshness -> rebase/reconcile/re-gate -> landing.

All three runs also emitted the same two stderr anomalies despite successful exit: a Terminated: 15 cleanup line and a missing temporary indep-blind/projects/blindproj/project.json. Repetition across all runs makes this deterministic or recurrent harness behavior, not a one-off. For a Trust Plane, noisy "success" evidence matters because future automated consumers must distinguish expected cleanup noise from infrastructure failure, test failure, and UNKNOWN.

Optimization principle

The next move should not be "run fewer tests." It should be more precise evidence invalidation: determine which changes invalidate which proofs, cache evidence against exact artifacts/revisions, shard independent checks, and reserve full-suite qualification for boundaries where it is actually required.

11. Semantic architecture evolution across 50 commits

The commit log confirms that the burst was broad architectural convergence, not just high activity. Exactly half of the 50 commits were fix(...) changes, 34% were docs(...), 12% were feat(...), and 4% were projects... changes. Only a small minority were explicit feature introductions; most landed work hardened semantics, encoded decisions, repaired edge cases, or institutionalized operating rules.

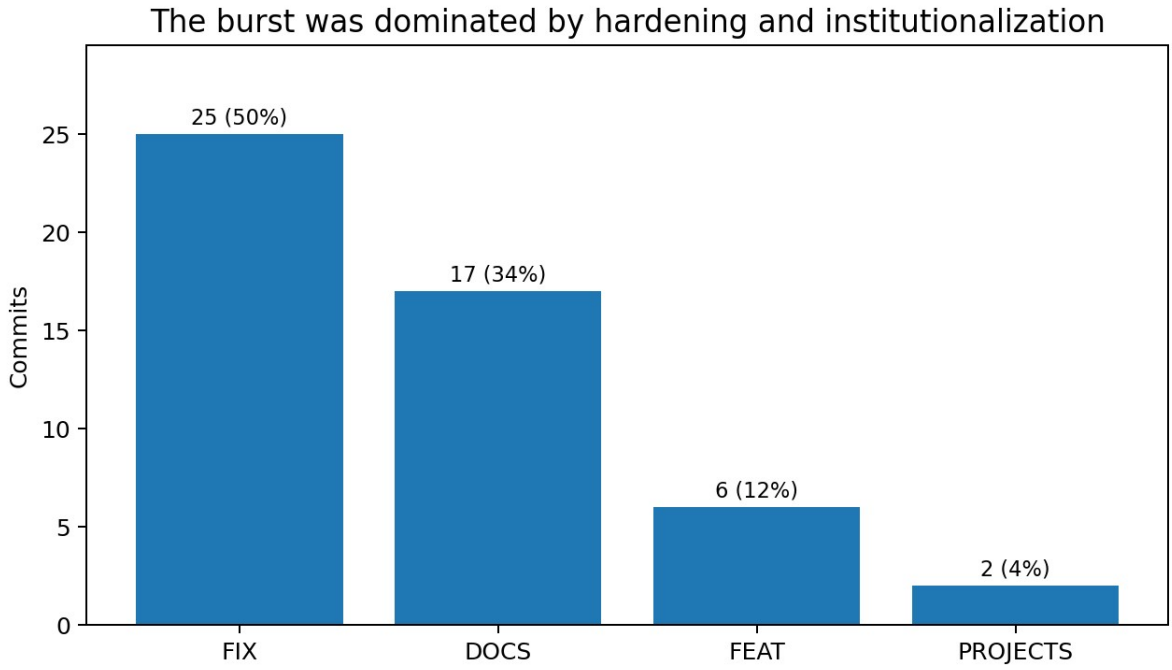


Figure 9. Commit-type composition: hardening and institutionalization dominated explicit feature work.

Using a single primary label per commit to avoid double-counting, the largest categories were Continuity (8), Trust/enforcement (7), Verification (6), Autonomy/unattended (6), Productization (5), Doctrine (4), Authority (3), Research/self-study (3), Concurrency/ownership (2), Observability (2), Capability recognition (2), Integration/landing (1), and Security/identity (1). Multi-theme commits exist; primary labeling is an analytical convenience, not a claim that each change affected only one dimension.

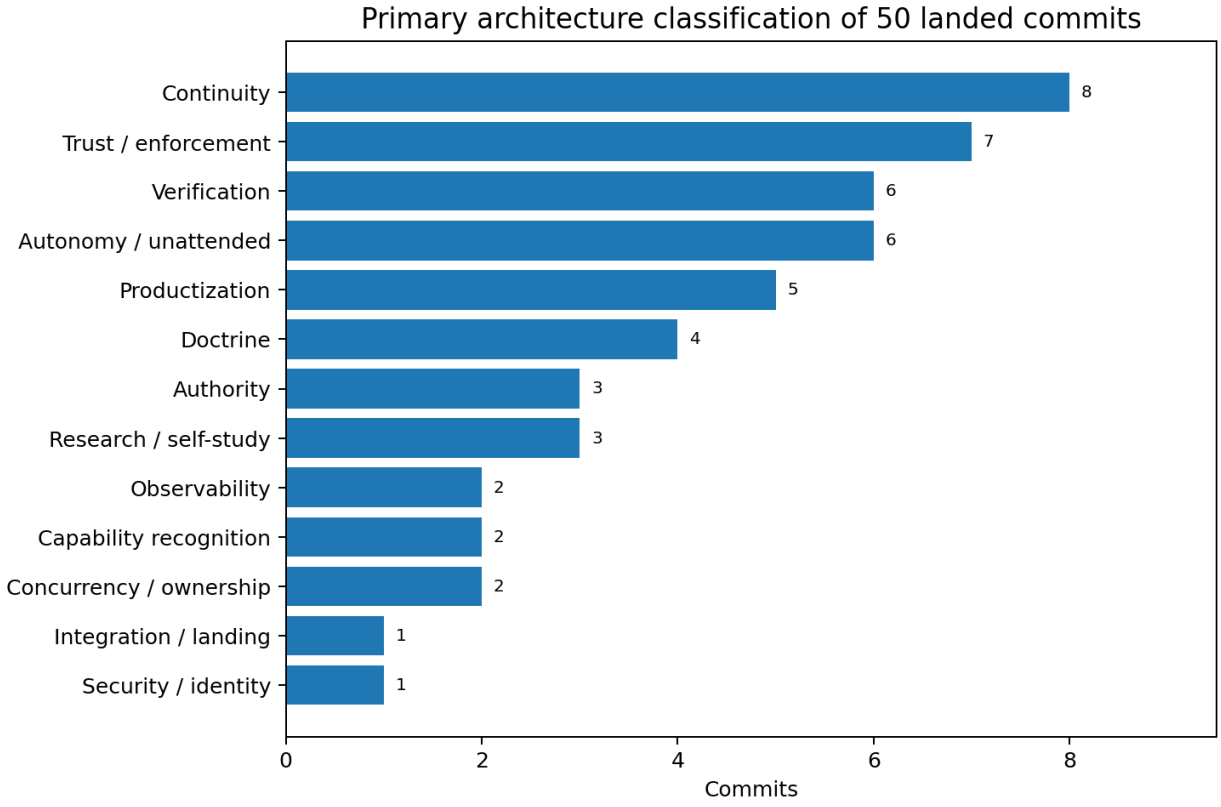


Figure 10. Primary architecture classification of all 50 landed commits.

11.1 Architecture sequence: institutional continuity becomes ownership

The continuity sequence begins with lifecycle logging: every start, resume, clear/compact, and Felix handoff becomes a durable row. HANDOFF.md then changes from one giant mutable document into a standing decision/current-state index whose sections live independently. Finally, the handoff architecture becomes concurrency-aware: each session writes its own file and a single fold actor assigns derived IDs, eliminating the shared-path collision mode. The sequence is not "more docs"; it is observability -> persistence -> ownership -> concurrent institutional memory.

11.2 Architecture sequence: verification becomes selective

The red.tsv and test commits show evidence semantics getting more precise. Published snapshots no longer fail merely because a recorded base is absent from repository history; docs-only or doctrine-only changes stop invalidating engine evidence; merge-base comparison prevents unrelated docs merges from staling every engine PR; and CPU-time assertions are corrected so host load does not produce false red. This is the beginning of dependency-aware evidence invalidation - the same direction independently suggested by the ~10-minute full-suite measurements.

11.3 Architecture sequence: deny evolves from strings to effects

The deny subsystem moves through several semantic layers: quoted messages/comments/here-doc content stop being mistaken for commands; bash -s heredoc scripts are correctly recognized as executable subject matter; large payloads are processed in one awk pass to avoid a 60s timeout; and finally deny rows can be scoped to the governed checkout while lib/where.sh reasons about where a command acts. The progression is string matching -> shell execution semantics -> performance under real payloads -> effect/location semantics.

11.4 Architecture sequence: autonomy becomes bounded institutionally

Unattended operation is progressively bounded rather than merely enlarged. Overnight authority belongs to the session that received it; repair/automerge reads the correct workflow and only proceeds on successful repair; unresolved checkouts and nothing-to-repair receive distinct exit semantics; repair acts only on what its ceiling counts; unattended templates pin the Felix version they consume; stale commission is explicitly human authority; background wake-ups cannot bypass stop checks; and the model is prohibited from reading /proc to copy its own API key into tracked changes. This is autonomy with explicit ownership, scope, exit states, identity, and authority boundaries.

11.5 Trust Plane versus Capability Plane

The 50-commit window is asymmetric. Trust, verification, continuity, authority, unattended safety, and institutional semantics advanced rapidly. By contrast, very little landed work corresponds to a mature standalone Capability Plane for task discovery, routing, dispatch, worker selection, or resource allocation. detect received portability/correctness work, but that is capability recognition rather than a full organizational capability scheduler.

Current asymmetry

Felix has a strong and fast-maturing Trust Plane/institutional substrate wrapped around substantial raw execution capacity. The explicit Capability Plane still lags. This matters because the end-state autonomous engineering organization needs both: one plane to decide and allocate work, another to prove, bound, and authorize what may become canonical.

12. Recursive improvement assessment

Calling any self-hosted developer tool "recursive" is easy; the stricter test is whether operation exposes problems, those problems are encoded as mechanisms, and the mechanisms change subsequent operation. Felix satisfies that test at a meaningful level.

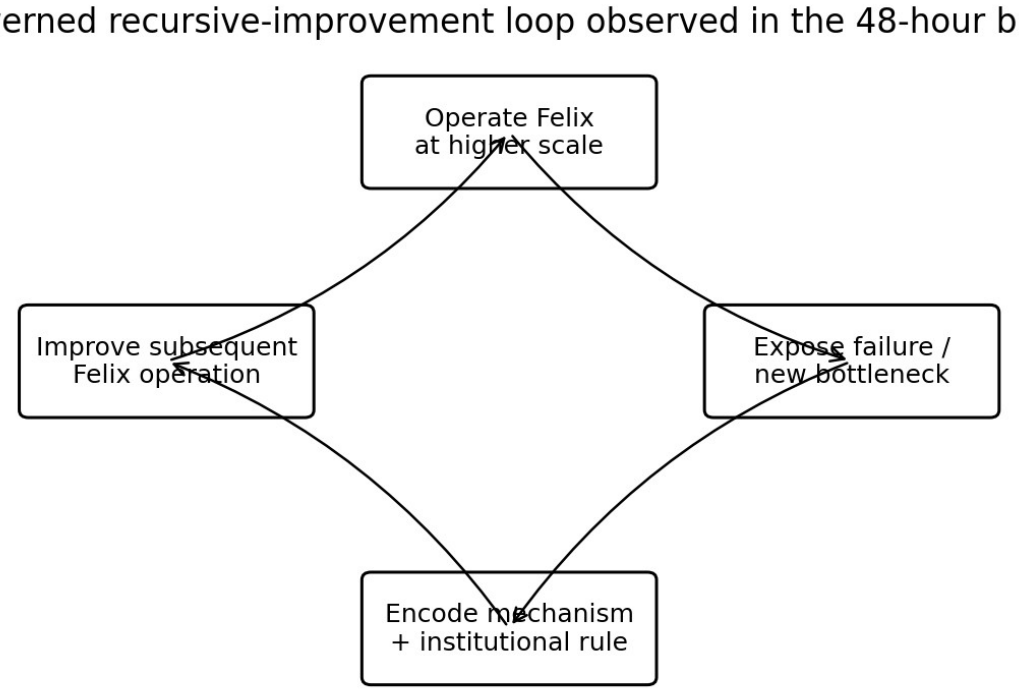


Figure 11. The governed recursive loop observed during the burst.

Level	Definition	Assessment
L0	AI helps the human write Felix.	Exceeded.
L1	Felix is used to develop Felix.	Clearly present at high scale.
L2	Operating Felix reveals its own problems and those become Felix mechanisms.	Clearly present: handoff ownership, audit, readiness/landing, evidence semantics.
L3	New mechanisms improve subsequent Felix development, concurrency, or reliability.	Supported by immediate reuse of continuity, ownership, verification, and governance mechanisms.
L4	The system systematically discovers bottlenecks, proposes/executes changes, and independently validates them.	Touched, not achieved. Claude task suggestions and Felix self-audit/research help, but human authorization and architectural judgment remain central.
L5	Near-unattended continuous recursive improvement.	Not supported by current evidence.

The strongest examples are causal rather than rhetorical. High-concurrency handoff collisions produced a per-session ownership/fold model. Candidate staleness produced the landing-actor requirement. Growing verification cost coincided with selective invalidation semantics. Running felix audit exposed both malformed-evidence robustness and subagent-visibility gaps. The forensic process itself repeatedly generated requirements that map directly onto the next Felix mechanisms.

What is not yet proven is acceleration of the recursive-improvement rate itself. A two-day burst can establish a recursive loop; it cannot establish that the derivative of improvement rate is increasing. Claims of runaway or autonomous recursive self-improvement would exceed the evidence.

13. What Felix is becoming - and what it is not yet

The best current description is: a human-directed, agent-executed, self-hosted recursive engineering system that is evolving toward an autonomous engineering institution. The word institution matters because Felix increasingly maintains durable state, authority boundaries, evidence tied to revisions, ownership rules, lifecycle records, and governed unattended behavior rather than simply invoking coding agents.

The system is not yet a fully autonomous engineering organization. The human still selects which suggested tasks deserve resources, resolves high-level ambiguity, makes direction-level decisions, clears certain authority states, decides which candidates are superseded or worth institutionalizing, and observes/coordinates the integration frontier. Those are not incidental leftovers; they define the current ceiling on autonomy.

Most important positive signal

Concurrency increased without an obvious collapse in verification discipline. The combination of 12+ concurrent sessions, near-parity engine/test growth, 64% of landed commits touching tests/run, 363 actual stop holds, and low textual shepherding per session is more meaningful than raw token volume or LOC.

14. Control UI and the human command plane

The next UI should not begin as a consumer-simple chat application. At the current stage, it should be a dense engineering-management console - closer to mission control, a Bloomberg terminal, or a kernel debugger - because Felix still requires the human to understand a wide set of organizational states. The organizing principle should be human action/state, not a flat list of agent sessions.

Primary queue	What it means	Typical human action
NEEDS AUTHORITY	A decision is outside delegated scope or changes a control.	Approve, reject, narrow, or redirect.
READY TO LAND	Evidence is current and candidate is qualified.	Authorize/observe landing or allow policy-driven auto-land.
STALE EVIDENCE	Candidate was qualified against an older relevant state.	Re-gate, reconcile, or supersede.
CONFLICT / SUPERSEDED	Parallel candidate overlap or newer work displaced it.	Choose winner, combine semantics, or retire.
BLOCKED	External dependency, missing fact, or failed prerequisite.	Supply fact/authority or defer.
BUG / INCIDENT	Execution exposed a new failure mode.	Escalate severity; potentially spawn follow-on work.
RUNNING	No human action currently needed.	Usually hide or collapse by default.

The hard architectural problem is preserving one-task-per-session isolation while allowing one singular UI to route decisions back to the correct session, branch, revision, and authority context. The UI must therefore sit over a machine-readable organizational state model, not scrape terminal text.

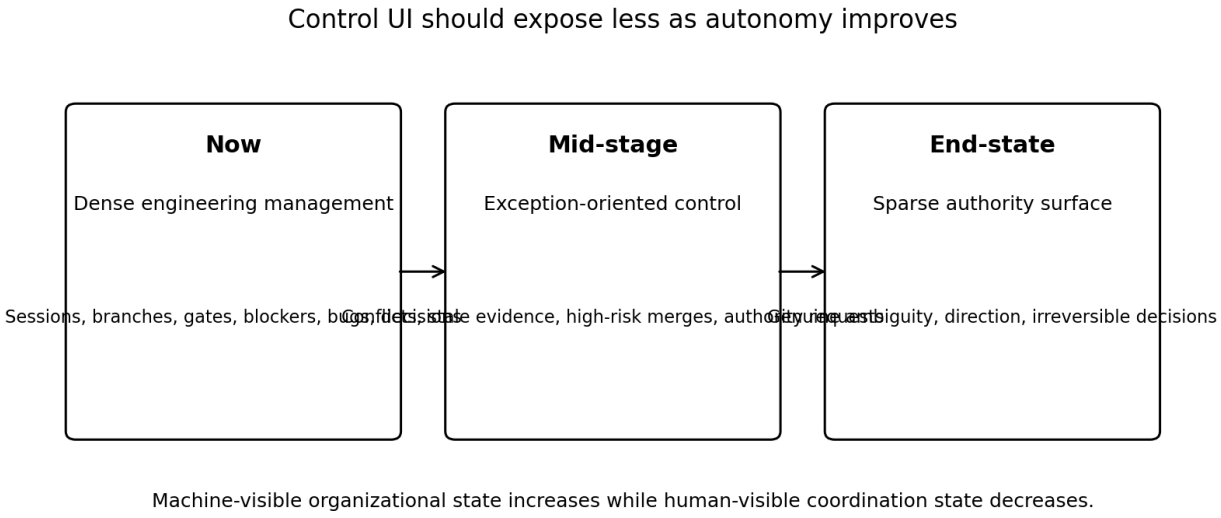


Figure 12. Desired evolution: the UI becomes quieter as machine autonomy increases.

14.1 Suggested Tasks are already a crude Capability-Plane primitive

Claude's in-session Suggested task cards demonstrate a useful pattern: the executing model discovers follow-on work but does not automatically expand scope; it proposes the work, the human authorizes a worktree, and a new isolated session executes it. Felix should preserve that bounded-authority shape while centralizing it.

A future task-intake queue should turn scattered suggestions into structured candidate work items containing source session, proposed objective, evidence/rationale, category, dependency graph, risk, expected independence, target repo/branch, and required authority. Low-risk work could eventually auto-spawn under an explicit grant; medium/high-risk work should remain visible in the command plane.

14.2 Lightweight external reference scan

A quick scan of current agent-development products supports the one-task/one-worktree plus central interface direction. Cursor documents an Agents Window in which agents run in isolated worktrees and are reviewed centrally. OpenHands Agent Canvas explicitly supports multiple agents in separate git worktrees visible from one interface. Factory likewise treats worktrees as first-class isolated session environments. These products validate the execution/UI skeleton; Felix can differentiate on organizational state, evidence freshness, authority, obligations, and landing semantics rather than merely showing many chats.

Reference	Reusable pattern	Felix-specific extension
Cursor Agents Window	Many agents across isolated worktrees; central review surface.	Organize by authority/evidence/integration state rather than by chat/session alone.
OpenHands Agent Canvas	Parallel agents, each isolated in a worktree, visible in one UI.	Add durable institutional state, qualification, candidate lifecycle, and human-action queues.
Factory App / Droid	Worktree-native sessions and parallel headless execution.	Bind execution to Trust Plane evidence, revision freshness, and landing authority.

15. Recommended next architecture

The next work should target the bottlenecks exposed by operating Felix at scale, not maximize the number of new primitives. The recommended order is intentionally biased toward preserving trust while increasing integration throughput.

1. Build the landing actor and explicit candidate lifecycle. Introduce machine-readable states such as ACTIVE, READY_FOR_GATE, GATING, NEEDS_REBASE, SUPERSEDED, ABSORBED, REJECTED, ABANDONED, and RESEARCH_ONLY. The landing actor should maintain a fresh landing frontier rather than periodically scan a pile of branches.

2. Make evidence invalidation dependency-aware. Extend the existing selective red/version semantics into an artifact/evidence graph: what changed, which proof it invalidates, which proof remains reusable, and which exact revision/artifact set the evidence covers.

3. Treat verification latency as a first-class budget. Shard independent checks, remove avoidable process sleeps/timeouts, cache exact-revision evidence, and measure gate critical path. Keep the full suite as a strong boundary, but stop charging it blindly to every candidate.

4. Expand organizational observability to subagents. More than half of raw token traffic is currently outside felix audit. The control plane should ingest parent sessions, subagents, branches, PRs, gates, evidence, lifecycle, dependencies, and authority requests into one event/state model.

5. Build the dense Control UI over that state model. Start with engineering states and action queues. Do not make a terminal manager with prettier tabs. Surface exceptions; collapse healthy running work; route decisions back to exact owner/session/branch/revision.

6. Productize Suggested Task intake. Turn model-discovered follow-ups into structured candidate work. Deduplicate them, link them to originating evidence, classify risk, and apply explicit auto-spawn policy rather than relying on scattered UI cards.

7. Fix trust-observability hygiene. Make audit tolerant of malformed evidence sources without losing all readable evidence. Remove recurring successful-suite stderr anomalies or encode them as explicit expected states.

8. Formalize the Capability Plane. After the Trust/continuity substrate stabilizes, make task discovery, routing, dispatch, worker selection, resource allocation, and capability discovery explicit rather than implicit in Claude UI behavior and manual spawning.

9. Productize this forensic report into Felix itself. A command such as felix organization status --since 48h should report generation throughput, verification throughput, integration throughput, human interventions, candidate age, trust holds, continuity, compute profile, external-project activity, and the current bottleneck.

16. Limitations and attribution caveats

- Raw token traffic is dominated by cache reads and should not be interpreted as 6.975B fresh inference tokens or a direct cost measure.
- The human-leverage analysis counts human text turns. Suggested-task approvals, Start with worktree clicks, window switching, and other UI actions are not captured as text turns.

- Felix audit does not see subagents, despite subagents representing 53.45% of measured raw token traffic. Its 35.7% stale/never figure applies only to the observable parent-session edit-stop sample.
- The audit recognizes a specific roster of checks and a subset of Bash write patterns; custom runners or unrecognized write forms can be missed.
- Lifecycle instrumentation landed during the observation period, so continuity counts are incomplete for the early window. Resume proves persistence, not correctness of restored institutional state.
- PR latency is reconstructed and right-censored. Slow PRs still open at the snapshot are absent from merged-latency statistics, making second-half latency estimates optimistic.
- Branch-ahead commit counts contain stacked histories, duplicated candidates, experiments, and superseded alternatives. The 175 global unique non-main commits are inventory, not 175 independent semantic changes.
- External worktree activity is not accepted output. BehaviorChangeBench was unborn, ReserveGrid contained stale/probe worktrees, and Elenchus had genuine ahead branches; these are not comparable with landed Felix commits.
- LOC and test-runner line counts are architecture/activity indicators, not productivity or correctness scores.
- The primary commit classification assigns one category per commit for counting. Many commits legitimately span multiple themes.
- Three test-runtime samples establish a useful local baseline, not a cross-machine performance benchmark.
- The evidence supports a recursive engineering loop, not autonomous runaway self-improvement or demonstrated acceleration of the improvement rate itself.

17. Conclusion

The 48-hour burst is significant because several independent evidence streams converge on the same story. Execution capacity is already abundant. Human textual shepherding per session is low. Verification grew nearly in step with engine code and actively blocked premature completion. Institutional continuity is no longer purely conceptual. At the same time, the system exposed a new coupled bottleneck: verification freshness, candidate reconciliation, landing, and human attention routing.

The architectural response is also becoming clear. Felix should not simply add more autonomous workers. It should make the organization itself explicit: task discovery and authorization, ownership, evidence, candidate lifecycle, authority, continuity, landing, and observability. The near-term Control UI is therefore a dense engineering command plane. If Felix succeeds, that UI should become progressively emptier - not because the system knows less, but because more organizational state becomes machine-managed and fewer situations genuinely require human authority.

The most important meta-result is that measurement itself is becoming part of the recursive loop. The act of auditing Felix uncovered malformed-input fragility, subagent invisibility, verification latency, candidate-aging economics, and the true human-attention distribution. Those observations are already specifications for the next Felix. The report is therefore not just a record of progress; it is a prototype of the observability/control surface Felix should eventually generate for itself.

Appendix A. Metrics snapshot

Metric	Measured value	Interpretation
Raw token traffic	6.975B	97.32% of total traffic was cache read; input-side cache-read share ~97.75%.

Metric	Measured value	Interpretation
Parent / subagent raw traffic	3.247B / 3.728B	Subagent share 53.45%.
Unique responses / tools	30,694 / 32,187	Bash 26,295 tool calls.
Active parent sessions	190	445 parent transcript files in human-turn scan.
Human text turns	471	40.2 assistant turns and 17.7 tools per human text turn.
<=1 / <=3 human text turns	81.6% / 85.3%	UI approvals are not included.
Code window	50 commits	162 files, +24,072/-6,929.
Engine / tests additions	5,995 / 6,060	Near parity in added lines.
Assertion registry	~2,735 -> 3,485	~+27.4%.
tests/run	~16,822 -> 21,988	~+30.7%.
Verification coupling	64% / 60%	Landed commits touching tests/run / assertions.tsv.
Audit edit stops	154	99 passed; 48 stale; 7 never; 0 failed.
Felix holds	363	Across 51 sessions.
Lifecycle	102 rows	53 startup, 32 resume, 16 handoff, 1 compact.
Resume share	37.6%	23/64 sessions resumed; 12/64 multi-day.
Full suite median	595.82s	3,511 passed, 0 failed, 3 skipped.
Active branches	77	175 globally unique non-main commits after dedup.
Open PR snapshot	~17	Integration/landing had become the new frontier.
External worktrees	29 total / 23 flagged active	Evidence of concurrent external activity, not landed output.

Appendix B. Commit-by-commit primary classification

Primary categories are used only to make the 50-commit burst countable. Several commits have obvious secondary themes (for example, snapshot is simultaneously productization, security, identity, and auditability).

SHA	Time	Primary category	Subject
d0b162f0	2026-09-22 04:28	Continuity	feat(lifecycle): every session start, resume, clear and compact, and every felix handoff, leaves one row in lifecycle.log — the record ECC's resume...
41f17bba	2026-09-22 11:12	Continuity	fix(route): the platform's summary request is its own session and is recorded as nothing, and Stop, SessionEnd and the ledger pass over it (#264) (...
4c65a974	2026-09-22 05:12	Productization	projects: govern the sv2-apps upstream clone
150e2bda	2026-09-22 11:21	Productization	projects(sv2-apps): drop the owner-step runbooks; email decision persists in git config
451bd401	2026-09-22 11:26	Continuity	docs(handoff): HANDOFF.md becomes the standing decision, the current state and an index; its 60 sections move verbatim to docs/handoff/<id>.md (#283)
091a8889	2026-09-22 13:47	Productization	feat(snapshot): the public snapshot is a command, with an audit that is not its scrub; secret-scan learns temporary AWS keys, and the unattended te...
f4b68c2b	2026-09-22 14:34	Observability	feat(audit): felix audit reports,

SHA	Time	Primary category	Subject
			read-only from a plain clone, how often a turn stopped on an edit nothing had checked; felix new --no-commission ...
437005ef	2026-09-22 15:28	Verification	fix(tests): the red.tsv base check skips where its base is no commit in this repository's history, as in a published snapshot, and still fails on a...
a5bcf61f	2026-09-22 16:34	Verification	fix(tests): the red.tsv and version checks apply only to a branch that changes the engine, so a docs-only or doctrine-only pull request passes agai...
2362602e	2026-09-22 16:43	Continuity	docs(handoff): §0br, the session of 2026-09-22 at close (#287)
b87b7827	2026-09-22 18:28	Productization	fix(commission): felix commission hashes the provider topology at the default IFS, so a checkout path with a space reads an answer the same way ses...
774f5a56	2026-09-22 19:00	Productization	docs(charter): Felix is a product for other people's projects, built first on Claude Code; the Phase-1-only scope is replaced (founder decision 202...
5c2fd878	2026-09-22 19:31	Capability recognition	fix(detect): felix_detect quotes the checkout root, so a path holding a space shows the capabilities its files do (#292)
9cd9f36f	2026-09-22 20:10	Doctrine	docs(roadmap): the build order to 2027-01-31 after Felix became a commercial product; it replaces direction-decisions §5's queue (#293)
f1174921	2026-09-22 20:30	Trust / enforcement	fix(escape): a renamed table lists the name it left, so renaming an enforcing table away fires the verifier (#291)
7dbb77d5	2026-09-22 20:47	Doctrine	docs(roadmap, handoff): build Felix first, commercialisation follows quality; §0bs, the session of 2026-09-22/23 (#295)
f224b0ca	2026-09-22 20:59	Authority	fix(qualify): qualification.tsv escapes to a person when a control is removed or changed, a control's verdict is fixed by its type, and a checker s...
eecfa506	2026-09-22 21:47	Capability recognition	fix(detect): the manifest probes glob against the checkout root, not the current directory, so felix doctor from a folder holding requirements/dev....
f781de0b	2026-09-22 22:29	Verification	fix(tests): red.tsv's base check compares the engine at the recorded base with the engine at the merge-base, so a docs-only merge to main no longer...
4f475c35	2026-09-22 23:11	Verification	fix(tests): the granted count in felix obligations' discovered/admitted sentence is asserted on every run, against a fixture that makes the sentenc...
5110ca64	2026-09-22 23:41	Trust / enforcement	feat(table): the table validator, schemas.tsv and the gate's schema check (queue item 2, step 1) (#298)
ca6389bf	2026-09-23 01:46	Concurrency / ownership	fix(facts): the fact log confirms its own rows, so a second gate appending to it at the same time

SHA	Time	Primary category	Subject
			no longer fails the probes check (#300)
c26feb7c	2026-09-23 02:57	Continuity	docs(handoff): §0bt, the table validator shipped (#291, #298) and deployed as 0.49.150 (#302)
47fe3978	2026-09-23 03:35	Trust / enforcement	fix(deny): a Bash row is refused only when it still matches the command the shell runs, so here-documents, quoted messages and comments stop being ...
580a2d2a	2026-09-23 04:09	Continuity	docs(handoff): §0bu, the session of 2026-09-23: the Elenchus candidates checked, deny rows read the command the shell runs, three Phase 0 items gat...
1de15917	2026-09-23 04:40	Trust / enforcement	fix(deny): a here-document handed to bash -s with arguments is still the shell's script, so the command in it is refused again (#313)
97557894	2026-09-23 05:04	Observability	fix(audit): a --verify that matches a backslash keeps a multi-line command's line breaks on macOS awk, where each became a backslash and joined the...
07bf6fea	2026-09-23 05:38	Trust / enforcement	fix(deny): a payload past a kilobyte has its subject read by one awk pass, so a large Bash command no longer runs the hook past its 60 s timeout (#...
9b8bdf0a	2026-09-23 05:53	Research / self-study	docs(episodes): the founder-Felix pipeline measured on twenty episodes, and why the interaction layer has to be mechanisms rather than advice (#316)
78ea7485	2026-09-23 06:01	Autonomy / unattended	fix(overnight): a night belongs to the session it was granted to, carries through the platform's chain on its own bounds, and counts a continuation...
b4798f57	2026-09-23 06:22	Autonomy / unattended	fix(repair, automerge): the repair ceiling and the auto-merge streak read the gate workflow's runs and no other's, and the unattended template proc...
685a5893	2026-09-23 06:34	Continuity	docs(handoff): §0bv, the night of 2026-09-23: overnight mode belongs to a session (#307), #303 carried to merge, #304 held for the founder's admiss...
945c0299	2026-09-23 11:36	Doctrine	docs(restart-moments): the step-1 scope — Felix names an engine the session is behind, records idle resumes, and defers the 500K restart behind the...
b40d82e0	2026-09-23 11:54	Authority	docs(constitution): direction-level documents wait for the founder's yes; nothing else waits (#319)
f1a883c6	2026-09-23 12:44	Autonomy / unattended	fix(repair): a checkout Felix cannot resolve exits 4, and the unattended template fails the job on it instead of escalating it as a spent ceiling (...
bd740748	2026-09-23 16:05	Doctrine	docs(roadmap, constitution): restart moments step 1a at the end of Phase 0 and 1b in Phase 1; the founder allows one line mid-session, narrowly (#322)
1375e6ee	2026-09-23 18:21	Autonomy / unattended	fix(repair): nothing to repair exits 5, not a brief's 0, and the

SHA	Time	Primary category	Subject
			unattended scope step reads repair's status under the bash -e Actions runs it in (...
9a0dfa4b	2026-09-23 19:00	Research / self-study	docs(episodes): batch 2 of the founder-episode harvest, and the session handoff (#327)
5692ff4a	2026-09-23 19:01	Autonomy / unattended	fix(repair): repair acts only on what its ceiling counts, an unreadable log is not a brief, and a failed install says why (#329)
7ac5315b	2026-09-23 19:21	Authority	docs(constitution): status reaches the founder in the session's closing notice, the monitor stays as the deeper page (decided 2026-09-23); the noti...
ff423008	2026-09-23 19:35	Verification	docs(tests): the issue-intake comment says what felix brief exits and how unattended.yml's mapping reads each code (#320)
8ccb7918	2026-09-23 20:00	Research / self-study	docs(handoff, episodes): the founder rated #332 "good." (yes) (#333)
685d37be	2026-09-23 20:12	Autonomy / unattended	fix(unattended): the template pins the Felix it checks out, so a copy and the engine it reads move together (#328)
b863a524	2026-09-23 20:39	Integration / landing	fix(merge): a stale commission receipt is the founder's to clear, not work Felix owes, and the advice follows the commission grant (#315)
541d3a5f	2026-09-23 21:20	Trust / enforcement	fix(stop): stop_hook_active ends a stop only when Felix's own hold started it, so the turn a background review wakes is checked, not waved through ...
3c960522	2026-09-23 21:34	Continuity	docs(handoff): §0by (#325's section, unchanged but for its id) and §0bz, restart moments scoped and placed (#326)
453f0006	2026-09-23 21:57	Verification	fix(tests): the 10,000 << shifts assertion times the audit's CPU, not the wall clock, so machine load cannot turn it red (#337)
f92b8ed7	2026-09-24 00:06	Concurrency / ownership	feat(handoff): a session adds its HANDOFF.md section as a file of its own, and one fold of main gives the ids, so no two sessions write the same pa...
0be007e1	2026-09-24 00:23	Trust / enforcement	feat(deny): a deny.tsv row may be scoped to the governed checkout, and lib/where.sh decides where a matched command acts, so a fixture under /tmp i...
c6989dc5	2026-09-24 00:40	Security / identity	fix(unattended): the model step disallows Read under /proc, so the API key in its own environment is not a file it can copy into a tracked change (...

Appendix C. Lightweight external UI references

These references were used only to sanity-check the control-UI/worktree direction. They are not evidence for Felix performance.

Source	URL	Relevant pattern
Cursor - Worktrees / Agents Window	https://prod.cursor.com/docs/configuration/worktrees	Isolated worktrees per task, centrally reviewed in the Agents Window.

Source	URL	Relevant pattern
Cursor - Multi-agent coding	https://prod.cursor.com/help/ai-features/multi-agent	Agents Window for managing many agents; async subagents and parallel execution.
OpenHands - Agent Canvas	https://www.openhands.dev/product/canvas	Parallel agents in separate git worktrees visible from one interface.
Factory - Worktrees in the Factory App	https://docs.factory.ai/factory-app/worktrees	First-class isolated worktree sessions for parallel tasks.
Factory - Droid CLI reference	https://docs.factory.ai/droid-cli/cli-reference	Native worktree flag and parallel headless sessions.